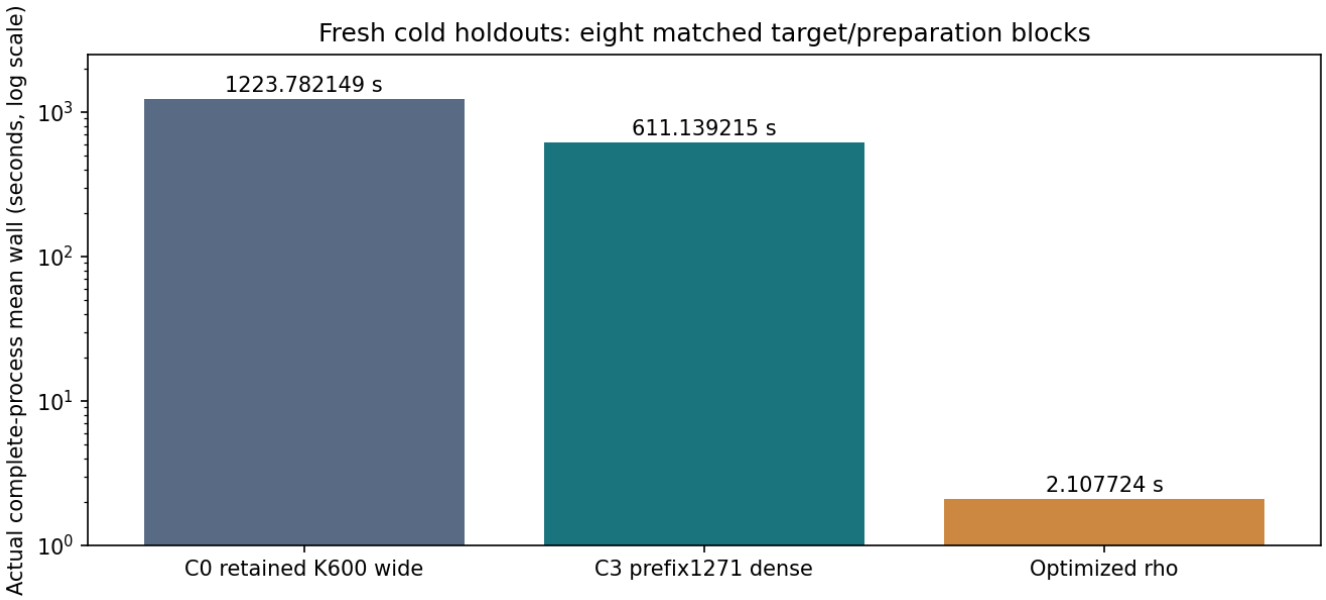
Degree-83 basis/setup: fresh cold holdouts

Stage started 2026-10-15 about18:42UTC. All26 declared processes completed and passed independent verification:26 canonical scalars,16 full IC rank/log audits and one identical rho A/A work pair; zero invalid/missing. This report is the cold stage, not completion of the full study.

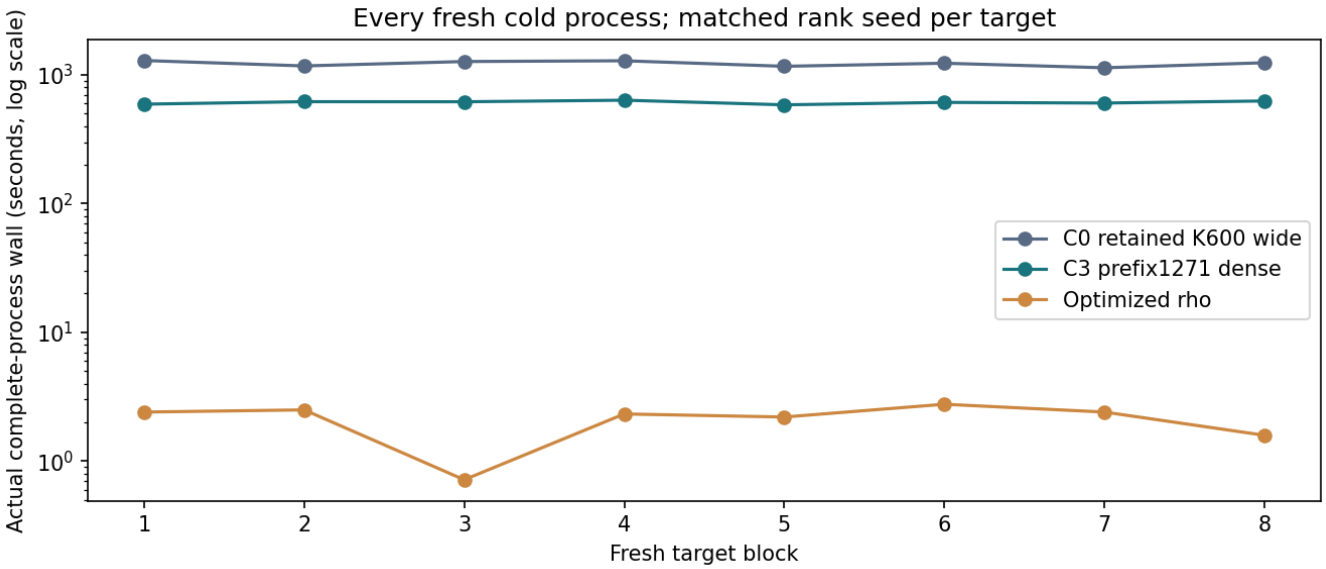
Observed complete-process means: selected C3 611.139215195s, K600 control1223.782148706s, optimized rho2.107724219s. All ratios below compare means of actual complete-process costs. Preparation is included.



Native comparison	Mean cost ratio	Conditional95% interval
C3/rho	289.952172	[245.967290, 375.396780]
C0/rho	580.617776	[489.180227, 758.610292]
C3/C0	0.499386	[0.484206, 0.514186]

C3 has lower measured cold cost than K600, but optimized rho has lower cold cost than both IC arms on this panel. For IC/rho, below1 favors IC; C3/C0 is explicitly an IC-basis comparison. No end-to-end IC-over-rho win is established.

Distribution and all target blocks



Role	Mean(s)	Median(s)	p90(s)	p95(s)	Max(s)
control	1223.782149	1235.837712	1287.738227	1289.737905	1291.737583
selected	611.139215	614.175880	629.222412	632.251110	635.279809
rho	2.107724	2.358451	2.573435	2.666188	2.758940

Native analysis uses midpoint conventional medians and type7 linear-interpolated quantiles. The20,000-draw paired percentile bootstrap uses seed202610157801 and resamples all eight matched target/preparation blocks. Intervals are conditional on this panel on one guest; they are not independent-host or population-wide guarantees.

Role	MeanCPU(s)	Mean aligned query(s)	MaxRSS(bytes)
control	1223.690379	1.355844	5834248192
selected	611.079537	0.217070	10832523264
rho	2.102676	1.878113	36573184

Aligned queries include generic public-Q validation, solve/recovery, generic scalar replay and answer serialization/write/flush in both methods. They exclude input-file loading and their own timing-receipt emission. Online comparisons remain secondary because actual preparation dominates this boundary. Memory-limit-hit remains null.

Every measured process, including control

Process	Wall(s)	CPU(s)	PeakRSS(bytes)
cold_0_control	1291.737583	1291.666394	5834174464
cold_0_selected	591.732233	591.685156	10832461824
cold_0_rho	2.398398	2.391504	36507648
cold_1_selected	618.685768	618.631645	10832388096
cold_1_rho	2.493933	2.490421	36507648
cold_1_control	1172.716256	1172.629776	5834248192
cold_2_rho	0.714578	0.713094	10948608
cold_2_control	1268.694905	1268.599469	5834166272
cold_2_selected	617.671766	617.612829	10832490496
cold_3_control	1286.024217	1285.924372	5834240000
cold_3_rho	2.318505	2.310324	36573184
cold_3_selected	635.279809	635.222192	10832519168
cold_4_rho	2.194436	2.190663	36368384
cold_4_selected	584.751779	584.683521	10832523264
cold_4_control	1164.867769	1164.756421	5834240000
cold_5_selected	610.679993	610.609521	10832388096
cold_5_control	1230.493906	1230.398680	5834240000
cold_5_rho	2.758940	2.754250	36503552
cold_6_control	1134.541037	1134.458637	5834108928
cold_6_selected	603.685988	603.627646	10832388096
cold_6_rho	2.399296	2.392685	36507648
cold_7_selected	626.626385	626.563787	10832449536
cold_7_rho	1.583708	1.578464	19439616
cold_7_control	1241.181517	1241.089280	5834174464
aa_3_0	1.842876	1.840532	19468288
aa_3_1	1.862018	1.856345	19464192

All26 completed; zero stage failures. Main rho retains253 internal capped walks in all costs. The A/A pair had identical deterministic work; larger/smaller process wall ratio1.010387. A/A time differences are not forced equal. Checker wall/CPU/RSS is separately metered per IC process and excluded from solver costs.

Scope, evidence and remaining study

Exact legacy curve: a=b=1, GF(2^83) polynomial z^83+z^7+z^4+z^2+1, prime subgroup r=8569786107849059 (53bits). Common native CPU3 reservation, oneOSthread,12GiB RLIMIT_AS and2100-second process cap. EPYC9V74 KVM guest; physical-host exclusivity and independent-host replication are not established.

C3 prefix1271dense was selected only on training and committed at19664f02fdcb5f3e5674f5a26e989a117be2f0bc before eight fresh holdouts were generated. Frozen remaining37process/100answer matrix and separate controller were published1e93e4ca1 before this stage. Rank seeds101/103/107/109/113/127/131/137 match across the two IC arms within each target block. Method order cycles prospectively.

Training retains two failed C4 allocations. Planned full scope is55processes/118answers; these are not actual verified totals. Eleven already-frozen cohort/A/A processes still follow:three independently seeded8-target shared-setup triplets plus the final A/A pair. Full-study analysis, final addendum, updated comprehensive PDF and evidence ZIP remain pending.

Independent native verification replays each canonical scalar and validates exact target order, seeds/policy/envelope, raw/matrix/binary identities, aligned-clock receipt boundaries and deterministic A/A work. Frozen independent checker verifies every IC anchor/four-point witness/intermediate, full modular row/rank gain, backsolve, representative[log]G and target log reconstruction. Exported probe/time counters are observations, not independently rerun search timings.

Original retained K600 construction and OS-cache flushing are unpriced. Endpoint VmRSS/HWM is not exclusive phase peak. No calibratedS, L2, independent-host, global optimum, asymptotic or deployed-security conclusion. The projected1685-target crossover remains unobserved and is not executed. Historical studies and the prior87-page PDF remain unchanged.

Evidence/source: ../../PROTOCOL.md

Evidence/source: native_statistics.json

Evidence/source: independent_receipt.json

Evidence/source: ../../stage_verification/check_stage.cpp

Evidence/source: ../../postprocessing/analyze_stage.cpp

Evidence/source: ../../postprocessing/statistics.hpp

PR https://github.com/aburan28/crypto/pull/1899 remains draft. No ledger promotion or merge; current CI and required human review remain gates. Python presents recorded native values and performs unit/format conversions only.